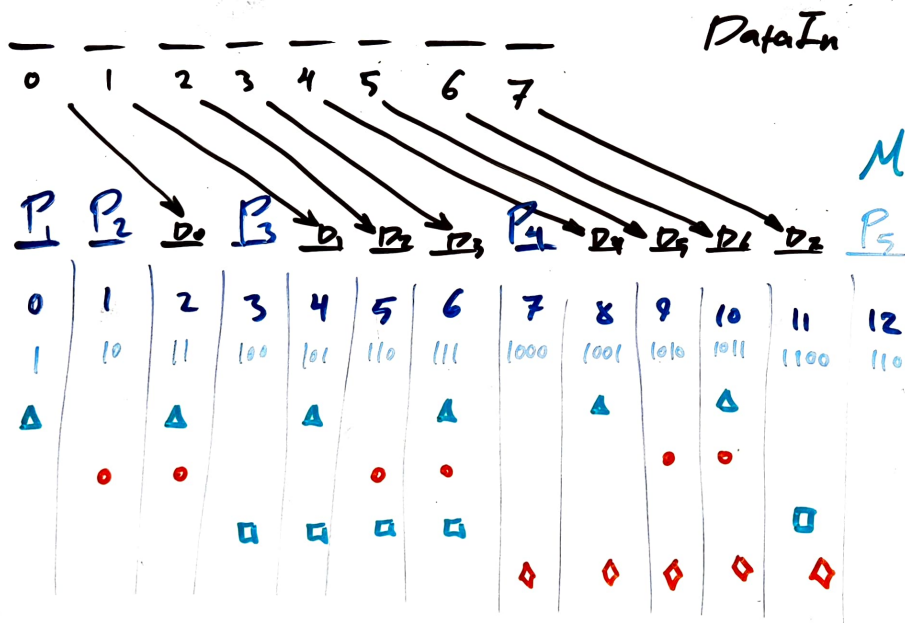


Encoder



Mode $\rightarrow$ if = 1 $\Rightarrow$ 100

Mode' $\oplus$ P $\Rightarrow$ 100

$$P_5 = \oplus 100$$

$$P_1 = D_0 \oplus D_1 \oplus D_3 \oplus D_4 \oplus D_6$$

$$P_2 = D_0 \oplus D_2 \oplus D_3 \oplus D_5 \oplus D_6$$

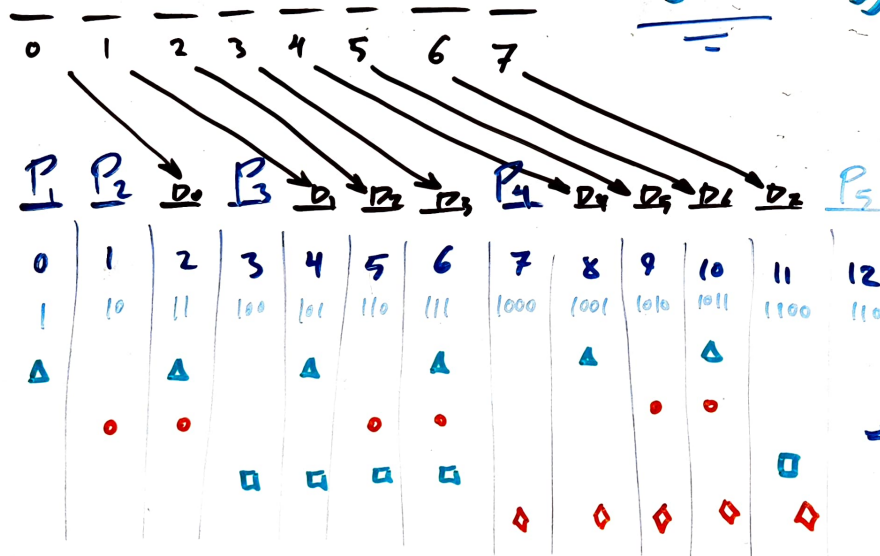
$$P_3 = D_1 \oplus D_2 \oplus D_3 \oplus D_7$$

if $\oplus D_i$ زنج $\Rightarrow P_k = 0$
حسابی - زنج

$$P_4 = D_4 \oplus D_5 \oplus D_6 \oplus D_7$$

Encoder

After Toggle if :



if P_5 was fine $\rightarrow$ OK ■

if P_5 wasn't fine

but $\overline{C_1 C_2 C_3 C_4} = 0$

$\rightarrow P_5$ was the error. ■

two bit error X

if P_i lie $\rightarrow q_i \oplus P_i = 1 \Rightarrow \text{check}_i = 1 \Rightarrow \overline{C_1 C_2 C_3 C_4} \rightarrow$ 1 base (1 base)

Mode-1 = 1: P_i should be 1 if $\text{check}_i = 1$

new Parity
if $q_i \oplus P_i = 1 \rightarrow$ lie

Encoder

New Parities = q

Data Out =

q_0, q_1, q_2, q_3

$$c_i = q_i \oplus p_i$$

$$0001 \quad I_0 \oplus c_0$$

$$0010 \quad I_1 \oplus c_1$$

$$0011 \quad I_2 \oplus (c_0 c_1)$$

$$0100 \quad I_3 \oplus c_2$$

$$0101 \quad I_4 \oplus (c_2 c_0)$$

$$0110 \quad I_5 \oplus (c_2 c_1)$$

$$0111 \quad I_6 \oplus (c_2 c_1 c_0)$$

$$1000 \quad I_7 \oplus c_3$$

$$1001 \quad I_8 \oplus (c_3 c_0)$$

$$1010 \quad I_9 \oplus (c_3 c_1)$$

$$1011 \quad I_{10} \oplus (c_3 c_1 c_0)$$

$$1100 \quad I_{11} \oplus (c_3 c_2)$$

Then if $p_4 = q_4 \Rightarrow$ Valid

if $p_4 \neq q_4$ & $c = 0000 \Rightarrow I_{12} \oplus 1 \Rightarrow$ Valid

Not Valid (Two Errors Happened.)